MTH1001-B

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

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Mathematical Structures

Module Leader: Professor Nigel Byott

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Use the Extended Euclidean Algorithm to find integers a and b such that $2023a + 960b = 1$. Hence find an integer x in the range $0 \leq x \leq 2022$ such that $960x \equiv 7 \pmod{2023}$. (12)

   (b) Let $\sigma = (134)(25)$ and $\tau = (137)(2468)$ in the symmetric group S_8. Express each of the permutations $\sigma\tau$, $\tau\sigma$ and $\sigma\tau^{-1}\sigma^{-1}$ as a product of disjoint cycles. Find the order and the sign of each of these permutations. (13)

   (c) Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 4 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 2 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}$$

   and

$$\mathbf{b} = \begin{pmatrix} 2 \\ 5 \\ 3 \\ 11 \end{pmatrix}.$$

   (i) Find a maximal linearly independent sublist of the list of vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$.

   (ii) Determine whether $\mathbf{b}$ can be written as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$ and, if it can, write it in this way. (13)

   (d) Let V be a vector space over $\mathbb{R}$. Show that if the vectors $\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n$ span V then the vectors $\mathbf{w}_1, 2\mathbf{w}_2, \ldots, n\mathbf{w}_n$ also span V. (12)

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SECTION B

2. (a) Let $n > 1$ be a positive integer, and let a, b, c, $d \in \mathbb{Z}$.

   (i) Give the definition of $a \equiv b \pmod{n}$. (2)

   (ii) Prove that if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ then $ac \equiv bd \pmod{n}$. (4)

   (iii) Prove that if $a \equiv b \pmod{n}$ then $na \equiv nb \pmod{n^2}$. (5)

   (b) Let $\mathbb{R}^\times$ be the group of nonzero real numbers (under the operation of multiplication) and let $\mathbb{R}$ be the group of all real numbers (under the operation of addition). What does it mean for a function $\theta : \mathbb{R}^\times \to \mathbb{R}$ to be a homomorphism of groups? (2)

   (c) In each of the following cases, determine whether the given function $\theta : \mathbb{R}^\times \to \mathbb{R}$ is a homomorphism of groups. Justify your answers.

   (i) $\theta(x) = x$ for all $x \in \mathbb{R}^\times$;

   (ii) $\theta(x) = 2\log|x|$ for all $x \in \mathbb{R}^\times$;

   (iii) $\theta(x) = \log|2x|$ for all $x \in \mathbb{R}^\times$;

   (iv) $\theta(x) = 0$ for all $x \in \mathbb{R}^\times$. (12)

   [25]

3. (a) Let $\mathrm{Fn}(\mathbb{R})$ be the vector space over $\mathbb{R}$ consisting of all functions $f : \mathbb{R} \to \mathbb{R}$. Determine whether each of the following subsets of $\mathrm{Fn}(\mathbb{R})$ is a subspace of $\mathrm{Fn}(\mathbb{R})$, briefly justifying your answers.

   (i) $\{f \in \mathrm{Fn}(\mathbb{R}) : f(x + 2\pi) = f(x) \text{ for all } x \in \mathbb{R}\}$;

   (ii) $\{f \in \mathrm{Fn}(\mathbb{R}) : f(x) \geq x \text{ for all } x > 0\}$;

   (iii) $\{f \in \mathrm{Fn}(\mathbb{R}) : f(2x) = f(x) \text{ for all } x > 0\}$. (11)

   (b) State the Fundamental Theorem of Arithmetic. (2)

   (c) Let $m > 1$, $n > 1$ be natural numbers such that $m^3 = n^2$. Show that there exists a natural number k with $m = k^2$ and $n = k^3$. (12)

   [25]

4. (a) List the elements of the group $\mathbb{Z}_{14}^{\times}$ of invertible elements in the integers modulo 14 (under the operation of multiplication mod 14). For each element $[a]$ of $\mathbb{Z}_{14}^{\times}$, find the order of $[a]$ and find the inverse of $[a]$. Is the group $\mathbb{Z}_{14}^{\times}$ cyclic? Is it abelian? (12)

   (b) Let V be a vector space over $\mathbb{R}$. Determine whether each of the following statements is true or false, and give a proof (if it is true) or a counterexample (if it is false).

   (i) if $\mathbf{u}$, $\mathbf{v} \in V$ then the list of vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u} - \mathbf{v}$ is linearly dependent;

   (ii) if the vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ span V then $\mathbf{u} \neq \mathbf{v}$;

   (iii) if $\phi, \psi : V \to V$ are functions such that the composite function $\phi \circ \psi : V \to V$ is a linear map, then ϕ is is a linear map. (13)

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